Mathematical Foundations of Signal Attenuation

1. Classical Energy Relations

The total kinetic energy in relativistic mechanics is given by the celebrated relation:

$$E = mc^2$$

For an electromagnetic field propagating through an absorbing medium with absorption coefficient alpha, the intensity I(x) at depth x satisfies the differential equation:

$$dI/dx = -\alpha * I(x)$$

Solving this relation yields the standard exponential decay law:

$$I(x) = I_0 * e^{-\alpha * x}$$

2. Discrete Fourier Transform

For a discrete sequence x_n of length N, the transform X_k is computed as:

$$X_k = \sum_{n=0}^{N-1} x_n * e^{-i * 2\pi * k * n / N}$$

where i = sqrt(-1).